THE MENTAL MODEL OF AN LLM – PART 2

Today's Agenda

- Reasoning Models vs Base Models
- Reading Model Evals & Benchmarks
- Comparing the Major Models (2026)
- Key Takeaways & Action Items

The 2025–2026 Split: Base Models vs Reasoning Models

Base Models

- Fast next-token prediction
- Lower cost & latency

Reasoning Models

- Uses extra compute to think step-by-step
- Much better at hard tasks

2024 (Base Era) → 2025–2026 (Reasoning Era)

What Are ‘Thinking Tokens’ Actually?

Internal Chain-of-Thought
Hidden reasoning process.

Self-Correction & Refinement
The model “talks to itself” before answering.

Billable Model Compute
You are charged for these internal steps.

High Volume vs. Output
Can be 3x–10x more than the final answer.

User Prompt → [Hidden Thinking Box] → Final Answer

Generation of Internal CoT.

When Should You Use Reasoning Models?

Task Type	Use Reasoning?	Recommendation
Simple chat/summarization	No	Base model
Hard math & science	Yes	High effort
Complex coding	Usually	Reasoning model
High volume work	No	Fast base

Reasoning Effort Controls (2026)

OpenAI o3

API `reasoning_effort`

Claude 4

`effort`

Best Practice: Start with Medium, escalate only when needed.

When Base Model + Good Prompting Wins

SCENARIOS WHERE BASE MODEL & PROMPTING EXCEL

1. Well-structured prompts with examples
2. Tasks with RAG or tools
3. Latency-critical applications
4. High-volume repetitive tasks
5. Tight budget scenarios

COMPARATIVE ANALYSIS: BASE VS REASONING

Metric	Base Model + Prompting	Reasoning Model
Cost	[Low to Medium]	[High / Effort-Based]
Latency	[Fast / Low]	[Slow / High]
Task Complexity	[Defined / Repeatable]	[Novel / Multi-Step]
Skill Requirement	[Prompt Engineering]	[Orchestration & Control]

Important Benchmarks in 2026

GPQA Diamond

Expert science

SWE-bench

Real software engineering

AIME / MATH

Advanced math

BFCL / Tau-bench

Agentic & tool use

LMSYS Arena

Human preference

Why Most Benchmarks Are Misleading

- Training data contamination
- Prompt sensitivity
- Leaderboard gaming
- Poor real-world correlation
- Saturation on easy tests

How to Read Leaderboards Skeptically

LMSYS Arena

(Human votes) – Most practical

Artificial Analysis

(Quality + Speed + Cost)

Your Own Micro-Eval

– Most Important –

Build Your Own Micro-Evaluation

1. Collect 20-50 real tasks from your work

2. Test models with identical prompts

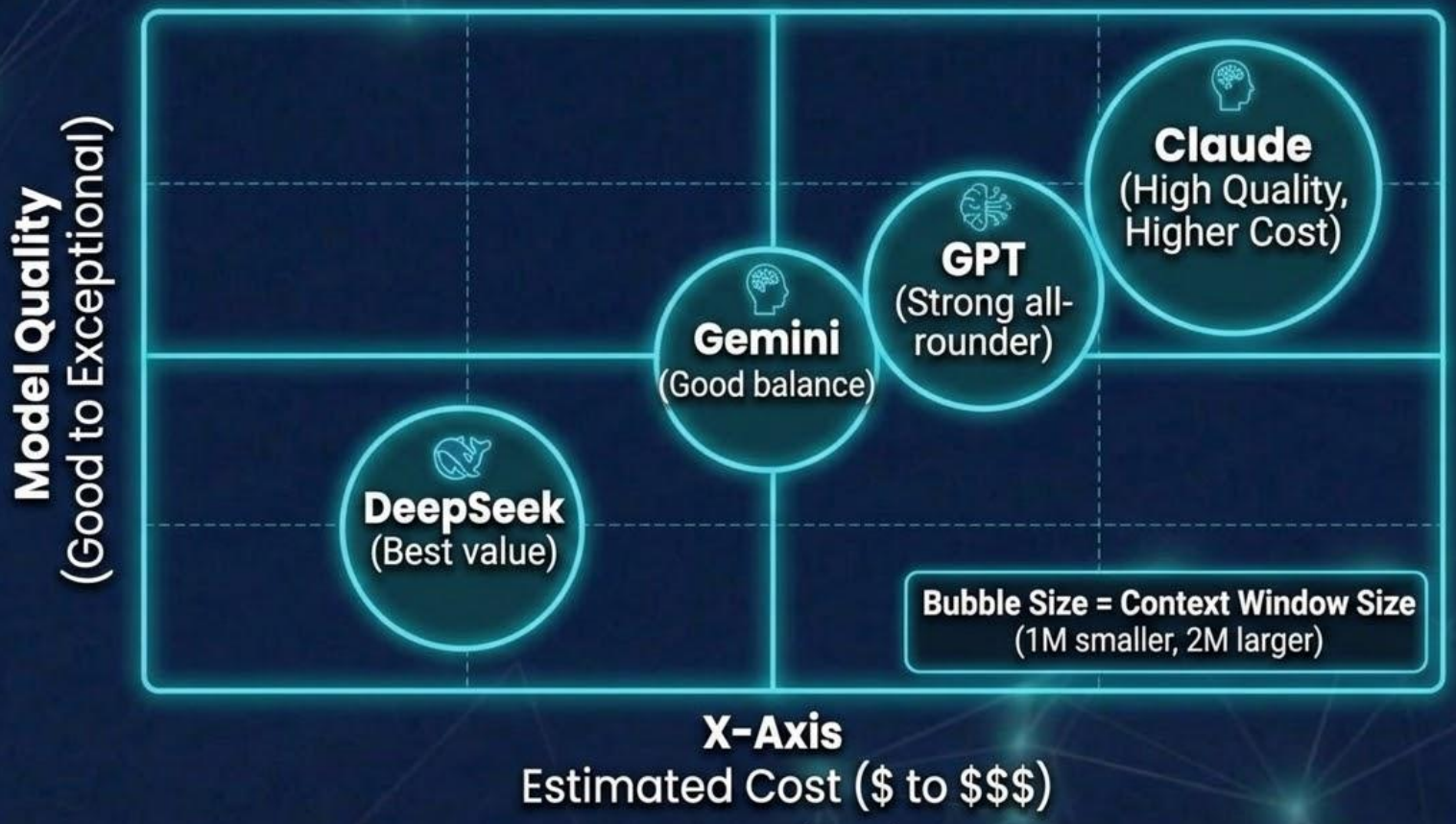
3. Score accuracy, quality, cost & speed

4. Repeat every 2-3 months

Major Models Comparison – 2026

Model Family	Reasoning	Coding	Writing	Context	Cost	Best For
OpenAI (o3)	Advanced ++	Expert	Eloquent	2M	$$$	Complex Reasoning / Long Horizon
Claude 4 Opus	Advanced ++	Expert	Eloquent	2M	$$$	Complex Reasoning / Long Horizon
Gemini 2.5	Advanced ++	Strong	Moderate	1M	High	Complex Coding / Strong Horizon
DeepSeek R1	Specialized	Strong	Concise	1M	$	Efficient Coding / Edge
Llama 4	Oprtialized	Strong	Concise	2M	$	Efficient Reasoning / Long Horizon

Cost vs Quality vs Context Tradeoffs



When Model Choice Really Matters

Matters A Lot

- High-stakes
- Complex agents
- Novel problems

Doesn't Matter Much

- Simple tasks
- Prototyping
- Casual use

Key Takeaways – Day 2

1. Reasoning models = paid test-time compute
2. Benchmarks are noisy — build your own evals
3. No single best model — context matters
4. Master effort knobs and routing

Action Items & Next Steps

1. Run a micro-eval this week
2. Implement effort routing
3. Track cost per task
4. Re-evaluate models quaterly

Thank You! Q&A

[Your Name / Team Name] [Email Address] [Company Website or Social URL]